

Flow–Field Equivalence of a Bridged Vessel Break: Schur–Complement Port Reduction and a Resistance–Fraction Tolerance

Methods note — thermoembolization 1D transport model

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Abstract

A segmentation-induced discontinuity in the extracted hepatic-artery centerline severs the 1D Hagen–Poiseuille flow network of Amare et al. and produces catastrophic false-negative predictions of bolus delivery. We formalize the alternative in which flow is routed from the inflow terminal, through the 3D tissue domain, and back into the outflow terminal. We show that, for any linear port-to-port bridge, the entire 3D block collapses via a Schur complement to a single effective conductance k_{br} , and that the bridged network reproduces the intact tumor flow *exactly* iff k_{br} equals the conductance of the missing segment. We then derive the tolerance around that exact condition: the fractional error in delivered flow equals the fractional error in k_{br} scaled by ϕ , the fraction of the inlet-to-tumor path resistance contributed by the broken segment. Because segment conductance scales as R^4 , ϕ is small for proximal/large-radius breaks and approaches unity for distal/small-radius breaks near the tumor, giving a computable, per-break map of when a break can be safely bridged and when accurate bridge calibration is required.

1 Governing 1D flow model (Amare et al.)

We adopt the reduced blood-flow model of Amare et al. verbatim. The extracted centerline is a graph in which each voxel is a pressure node and each connection a Hagen–Poiseuille element. For the element joining nodes i and j ,

$$q_{ij} = k_{ij} (P_i - P_j), \quad k_{ij} = \frac{\pi R_{ij}^4}{8\mu L_{ij}}, \quad (1)$$

where R_{ij} , L_{ij} are the radius and length of the element and μ the blood viscosity. Mass balance at every interior node i reads

$$\sum_{j \in N_i} k_{ij} (P_i - P_j) = 0, \quad (2)$$

a Dirichlet inlet condition fixes the root pressure,

$$P_{\text{root}} = P_{\text{MAP}}, \quad (3)$$

and each terminal node carries a Neumann/sink condition coupling it to the unsegmented bed at central venous pressure through a lumped conductance γ_a/μ ,

$$\sum_{j \in N_i} k_{ij} (P_i - P_j) + \frac{\gamma_a}{\mu} (P_{\text{CVP}} - P_i) = 0. \quad (4)$$

Assembling (1)–(4) over all nodes gives the linear system

$$L \mathbf{P} = \mathbf{f}, \quad (5)$$

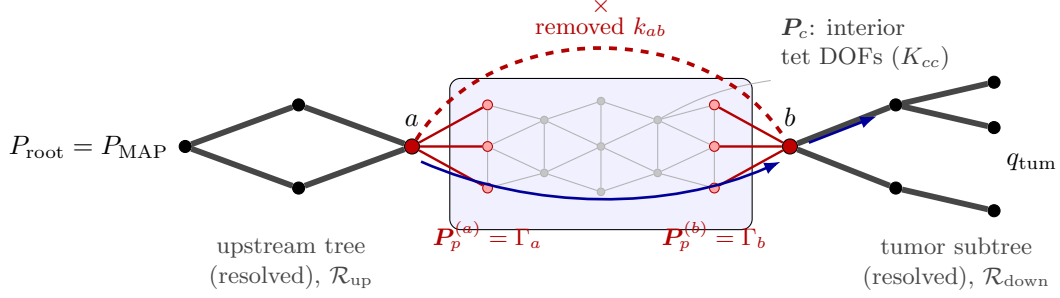


Figure 1: Break geometry underlying the block system (7). The direct Hagen–Poiseuille element of conductance k_{ab} (dashed, crossed) is lost to the segmentation break. The 3D bridge is a tetrahedral mesh: its interior vertex DOFs $\mathbf{P}_c$ (grey, coupled by K_{cc}) and its two *interface* DOF sets $\mathbf{P}_p = (\mathbf{P}_p^{(a)}, \mathbf{P}_p^{(b)})$ on the coupling faces Γ_a, Γ_b (red)—generally many nodes per face, not one. The 1D vessel connects to each face as a single lumped port a or b (red edges), so recovering the two-DOF port vector of §2 requires, in addition to eliminating $\mathbf{P}_c$ (Eq. (8)), an interface-lumping constraint reducing each Γ to one pressure (Remark 1). Flow enters at a , traverses the mesh, and returns at b to the resolved tumor subtree carrying q_{tum} ; the resolved upstream and downstream trees contribute $\mathcal{R}_{\text{up}}$ and $\mathcal{R}_{\text{down}}$ of §4.

where L is a weighted graph Laplacian (symmetric positive definite once the Dirichlet condition is imposed) with off-diagonal entries $-k_{ij}$, and $\mathbf{f}$ encodes the boundary data. All quantities of interest—in particular the flow delivered toward a tumor bed—are linear functionals of the solution $\mathbf{P}$ of (5). This linearity is the sole structural fact the results below require.

2 The break and its 3D bridge

Let a single element between an *inflow* node a and an *outflow* node b be lost to a segmentation break. In the intact network the two nodes are joined by the Hagen–Poiseuille conductance

$$k_{ab} = \frac{\pi R_{ab}^4}{8\mu L_{ab}}, \quad (6)$$

and the subtree feeding the tumor hangs off b . We denote by q_{tum} the total flow crossing b toward that subtree.

When the element is removed, we reconnect a and b through the 3D domain: the inflow terminal acts as a source into a porous/Darcy tissue block and the outflow terminal as a sink drawing from it. Provided the 3D transport is *linear* (Darcy, Stokes, or a further Hagen–Poiseuille sub-network—valid before embolization-driven viscosity feedback sets in; see Remark 3), the 3D block is a linear two-port probed only at a and b .

2.1 Schur-complement reduction of the 3D block

Partition the degrees of freedom of the 3D bridge (Figure 1) into the *port* DOFs $\mathbf{P}_p$ and the *interior* DOFs $\mathbf{P}_c$. For a tetrahedral discretization the interior set $\mathbf{P}_c$ is the (large) collection of mesh-vertex pressures coupled by the internal stiffness K_{cc} , while the port set $\mathbf{P}_p = (\mathbf{P}_p^{(a)}, \mathbf{P}_p^{(b)})$ collects the interface vertices on the two coupling faces Γ_a, Γ_b where the 1D vessel meets the tissue—in general

$|\mathbf{P}_p| = |\Gamma_a| + |\Gamma_b| \gg 2$, not a single node per port (see Remark 1). The discrete linear operator (Darcy stiffness / internal Laplacian) has the block form

$$\begin{pmatrix} K_{pp} & K_{pc} \\ K_{cp} & K_{cc} \end{pmatrix} \begin{pmatrix} \mathbf{P}_p \\ \mathbf{P}_c \end{pmatrix} = \begin{pmatrix} \mathbf{g}_p \\ \mathbf{0} \end{pmatrix}, \quad (7)$$

where the interior right-hand side is zero because sources enter only at the ports. Eliminating the interior unknowns, $\mathbf{P}_c = -K_{cc}^{-1} K_{cp} \mathbf{P}_p$, yields the *Schur complement* of K_{cc} ,

$$S = K_{pp} - K_{pc} K_{cc}^{-1} K_{cp} \in \mathbb{R}^{|\mathbf{P}_p| \times |\mathbf{P}_p|}, \quad S \mathbf{P}_p = \mathbf{g}_p, \quad (8)$$

a dense face-to-face conductance matrix over all interface DOFs. The 1D network resolves each face only as a single lumped pressure and net flux, so we impose an interface-lumping constraint— $\mathbf{P}_p^{(a)} \equiv P_a \mathbf{1}$, $\mathbf{P}_p^{(b)} \equiv P_b \mathbf{1}$ (equal pressure across each face, with the face flux as its conjugate; Remark 1)—which reduces S to the 2×2 port operator $\hat{S}$. For a two-port with no internal sources $\hat{S}$ has the conductance (weighted-Laplacian) structure

$$\hat{S} = \begin{pmatrix} k_{\text{br}} & -k_{\text{br}} \\ -k_{\text{br}} & k_{\text{br}} \end{pmatrix}, \quad k_{\text{br}} \equiv (\hat{S})_{aa} = -(\hat{S})_{ab} > 0, \quad (9)$$

so that the port relation collapses to a single scalar,

$$\boxed{q_{a \rightarrow b}^{\text{bridge}} = k_{\text{br}} (P_a - P_b)}. \quad (10)$$

Remark 1 (What $\mathbf{P}_p$ is, and why lumping is a modeling choice). *The port set $\mathbf{P}_p$ is not two scalars in a tetrahedral bridge: it is the set of interface vertices on the coupling faces Γ_a, Γ_b , and the interior elimination (8) produces a dense $|\mathbf{P}_p| \times |\mathbf{P}_p|$ face-to-face conductance, not the 2×2 of (9). The single scalar k_{br} emerges only after a second reduction that lumps each face to one DOF. The lumping rule—uniform face pressure, flux-weighted average, or a single-node coupling—is a modeling choice that changes k_{br} , exactly the identifiability question already present in the terminal γ_a closure of (4), now posed at an interior interface. All results below hold once a lumping is fixed; the sensitivity of k_{br} to that choice is itself worth reporting.*

Proposition 1 (Complete port characterization). *However complex the internal geometry, permeability field, or discretization of the 3D bridge, its effect on the surrounding 1D network is fully described by the single effective conductance k_{br} of (9). All of gap length, gap width, tissue permeability, and bridge topology enter the global flow solve only through k_{br} .*

Consequently the bridged network is identical to a network in which a and b are joined by one Hagen–Poiseuille-like element of conductance k_{br} . The break has been reduced to a single edge weight.

3 Exact flow equivalence

Compare two networks that are identical everywhere—same upstream tree, same downstream tumor subtree, same terminal γ_a couplings—except for the single edge between a and b : the intact network carries weight k_{ab} (6), the bridged network carries weight k_{br} (9).

Proposition 2 (Exact equivalence). *Let $q_{\text{tum}}(k)$ denote the tumor-directed flow when the a – b edge has conductance k , all else fixed. Then q_{tum} is a rational (Möbius) function of k , and*

$$q_{\text{tum}}^{\text{broken}} = q_{\text{tum}}^{\text{intact}} \iff k_{\text{br}} = k_{ab} = \frac{\pi R_{ab}^4}{8\mu L_{ab}}. \quad (11)$$

Proof. Changing a single edge weight is a rank-one modification of the Laplacian in (5): $L(k) = L_0 + k \mathbf{e}_{ab} \mathbf{e}_{ab}^\top$, where $\mathbf{e}_{ab} = \mathbf{u}_a - \mathbf{u}_b$ is the incidence vector of the edge. By the Sherman–Morrison formula the solution $\mathbf{P}(k) = L(k)^{-1} \mathbf{f}$ is a componentwise Möbius function of k , hence so is any linear functional $q_{\text{tum}}(k)$. A Möbius function is injective on the physical branch $k > 0$ (its derivative does not vanish), so $q_{\text{tum}}(k_1) = q_{\text{tum}}(k_2)$ with $k_1, k_2 > 0$ forces $k_1 = k_2$. Taking $k_1 = k_{\text{br}}$, $k_2 = k_{ab}$ gives (11). $\square$

Remark 2 (Calibration target: unresolved lumen, not bulk tissue). Equation (11) states the bridging requirement precisely: the 3D bridge must reproduce the conductance of the vessel segment it replaced. A break is almost always unresolved lumen rather than parenchyma, so k_{ab} is a high-conductance channel. Bridging instead with bulk-tissue permeability gives $k_{\text{br}} \ll k_{ab}$, starving the downstream tumor and reintroducing the false-negative. The correct bridge is therefore a vessel-like channel whose port-reduced conductance matches k_{ab} —the interior analogue of the terminal γ_a closure in (4), now applied to an interior gap.

4 Tolerance: the resistance-fraction ϕ

Exact equality requires the true k_{ab} , which is unknown precisely where the vessel broke. The operational question is therefore how far k_{br} may deviate from k_{ab} before the tumor flow changes appreciably. Reduce the inlet-to-tumor path to three conductances in series: the port-reduced upstream resistance $\mathcal{R}_{\text{up}}$ seen at a , the broken segment $1/k$, and the port-reduced downstream resistance $\mathcal{R}_{\text{down}}$ seen from b into the tumor bed. The delivered flow is

$$q_{\text{tum}}(k) = \frac{\Delta P}{\mathcal{R}_{\text{tot}}(k)}, \quad \mathcal{R}_{\text{tot}}(k) = \mathcal{R}_{\text{up}} + \frac{1}{k} + \mathcal{R}_{\text{down}}, \quad (12)$$

with ΔP the effective driving pressure across the path. Define the *resistance fraction* of the broken segment in the intact network,

$$\phi = \frac{1/k_{ab}}{\mathcal{R}_{\text{tot}}(k_{ab})} = \frac{1/k_{ab}}{\mathcal{R}_{\text{up}} + 1/k_{ab} + \mathcal{R}_{\text{down}}} \in (0, 1). \quad (13)$$

Two points on this definition. First, ϕ is built from the *true* segment conductance k_{ab} and the *intact* total resistance $\mathcal{R}_{\text{tot}}(k_{ab})$ —not from the bridge k_{br} —because ϕ is a property of the anatomy being reproduced, the fixed reference against which any bridge error is judged. It answers “how much leverage does this location have over the delivered flow?” and that leverage is a fact about the vessel that should be true, evaluated as if the segment were still present, independent of whatever bridge one later inserts. Using k_{br} here would make ϕ move every time the bridge changed and would forfeit exactly the property that makes it useful: computability from the resolved network *before* any bridge value is chosen (Corollary 1). Second, the two conductances play distinct roles that the tolerance result (14) keeps separate. k_{ab} sets the *reference*—where the flow should land—and enters only through ϕ , the *sensitivity*; k_{br} is the *controllable variable*, and enters only through its relative deviation $\delta = (k_{\text{br}} - k_{ab})/k_{ab}$. The product $\phi \delta$ then factors cleanly into “how sensitive is the flow to a bridge error here” (ϕ , from k_{ab}) times “how large is my bridge error” (δ , from k_{br}). Formally, ϕ is the sensitivity $\partial q_{\text{tum}} / \partial (1/k)$ evaluated at the exact-equivalence operating point $k = k_{ab}$ of Proposition 2, since a first-order tolerance is a statement about deviations *away from* the correct value—and the correct value is k_{ab} .

Proposition 3 (First-order flow tolerance). *Let the bridge conductance carry a relative error $\delta = (k_{\text{br}} - k_{ab})/k_{ab}$. Then to first order the relative error in delivered tumor flow is*

$$\frac{q_{\text{tum}}^{\text{broken}} - q_{\text{tum}}^{\text{intact}}}{q_{\text{tum}}^{\text{intact}}} \approx \phi \frac{\delta}{1 + \delta} \approx \phi \delta \quad (|\delta| \ll 1). \quad (14)$$

Proof. From (12), $q_{\text{tum}} \propto \mathcal{R}_{\text{tot}}^{-1}$. Perturbing $k: k_{ab} \rightarrow k_{\text{br}}$ changes only the middle term, $1/k_{ab} \rightarrow 1/k_{\text{br}} = (1/k_{ab})/(1 + \delta)$, so

$$\mathcal{R}_{\text{tot}}(k_{\text{br}}) = \mathcal{R}_{\text{tot}}(k_{ab}) - \frac{1}{k_{ab}} \frac{\delta}{1 + \delta}.$$

Dividing by $\mathcal{R}_{\text{tot}}(k_{ab})$ and using (13), $\mathcal{R}_{\text{tot}}(k_{\text{br}})/\mathcal{R}_{\text{tot}}(k_{ab}) = 1 - \phi \delta / (1 + \delta)$, and since $q_{\text{tum}}^{\text{broken}}/q_{\text{tum}}^{\text{intact}} = \mathcal{R}_{\text{tot}}(k_{ab})/\mathcal{R}_{\text{tot}}(k_{\text{br}})$, inverting and linearizing gives (14). $\square$

The interpretation is the operational payoff. The delivered-flow error is the bridge-conductance error *attenuated* by ϕ :

- $\phi \ll 1$ (**wide tolerance**). When the broken segment contributes little of the path resistance, even a large mis-estimate of k_{br} barely moves the tumor flow. Because $1/k_{ab} \propto L_{ab}/R_{ab}^4$, this is the regime of *proximal, large-radius* breaks: the path resistance is dominated by the small distal vessels downstream, which the break leaves untouched. Here a break may be bridged crudely—or even left as any reasonable high-conductance channel—without harming the delivered-dose prediction.
- $\phi \rightarrow 1$ (**narrow tolerance**). When the broken segment is itself the dominant resistance—*distal, small-radius* breaks near the tumor, where R_{ab}^4 is smallest—the tumor flow tracks k_{br} almost one-for-one and the bridge must be calibrated accurately to k_{ab} .

Corollary 1 (Per-break tolerance map, computable without the gap). *$\mathcal{R}_{\text{up}}$ and $\mathcal{R}_{\text{down}}$ are Schur-reduced resistances of the resolved network at a and b and are obtained from the intact solve; k_{ab} enters ϕ only through the assumed gap geometry. Hence ϕ can be estimated for every candidate break from the surrounding resolved vasculature, yielding an a priori map of which breaks fall in the wide- versus narrow-tolerance regime before any bridging is performed.*

5 Worked example: proximal versus distal break

To make the tolerance concrete we take a single three-generation path from the injection site to a tumor terminal (Remark 4), with radii tapering from a lobar artery to a subsegmental feeder and roughly constant segment length. Radii and lengths, and the resulting Hagen–Poiseuille conductances (6), are listed in Table 1. The downstream tumor-bed closure is set to 40% and the proximal (aorta/celiac) feed to 5% of the summed segment resistance, so that—as in vivo—the small distal vessels plus bed carry most of the path resistance.

Figure 2 shows the exact flow ratio $q_{\text{tum}}^{\text{broken}}/q_{\text{tum}}^{\text{intact}} = \mathcal{R}_{\text{tot}}(k_{ab})/\mathcal{R}_{\text{tot}}(k_{\text{br}})$ as the bridge conductance is swept away from the true value. Two readings make the point. First, bridging the distal break with a near-tissue conductance ($\delta \rightarrow -1$, i.e. $k_{\text{br}} \ll k_{ab}$) collapses the delivered flow to 0.16 of its intact value—the false-negative of Remark 2 rendered quantitatively—whereas the same crude bridge leaves the proximal break at 0.91. Second, the first-order estimate (14) matches the exact curve closely in the wide-tolerance regime (e.g. $\delta = +0.5$ gives exact 1.0037 vs. predicted 1.0037 for g1) and remains a usable guide even for the distal break (1.251 exact vs. 1.200 predicted), where the Möbius curvature of Proposition 2 begins to show.

segment	R (mm)	L (mm)	k_{ab} (SI)	ϕ	δ for 10% flow error
g1 (lobar, proximal)	1.20	15	1.55×10^{-8}	0.011	$\pm 900\%$
g2 (segmental)	0.70	12	2.25×10^{-9}	0.077	$\pm 130\%$
g3 (subsegmental, distal)	0.40	10	2.87×10^{-10}	0.602	$\pm 17\%$

Table 1: Three-generation delivery path. A break in the proximal large-radius segment (g1) contributes almost none of the path resistance ($\phi = 0.011$) and so tolerates an enormous bridge-conductance error; a break in the distal small-radius feeder (g3) is itself the dominant resistance ($\phi = 0.602$) and tolerates only a narrow error band. The $81\times$ spread in ϕ tracks the $(1.20/0.40)^4 = 81$ ratio of the fourth powers of the radii.

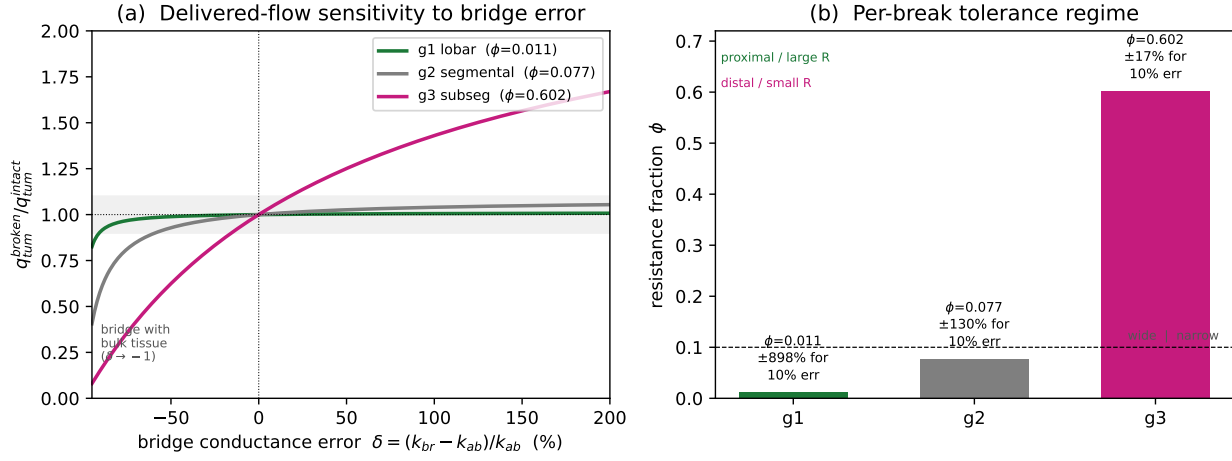


Figure 2: (a) Exact delivered-flow ratio versus bridge-conductance error δ for a break in each generation; the shaded band marks a $\pm 10\%$ flow tolerance. The proximal curve (g1) is nearly flat—delivered flow is almost independent of how the break is bridged—while the distal curve (g3) tracks δ steeply. (b) The resistance fraction ϕ by generation; annotations give the bridge error tolerated for a 10% flow error. Breaks with ϕ below the dashed line lie in the wide-tolerance regime and may be bridged crudely; those above require accurate calibration of k_{br} to k_{ab} .

6 Consequence for delivered acid chloride

The bolus saturation obeys the 1D advection–reaction transport of Amare et al.,

$$\frac{\partial s_o}{\partial t} + v \frac{\partial s_o}{\partial x} = -\Gamma(s_o, \epsilon), \quad (15)$$

advected at the velocity set by the network flow and with a *local* hydrolysis sink Γ . Past the outflow node b the advecting field is q_{tum} and the reaction term is unchanged by the break. Therefore preservation of q_{tum} to tolerance (14) implies preservation of the delivered-saturation (and hence delivered acid-chloride) profile past b to the same first-order tolerance $\phi\delta$. The flow-level equivalence is inherited at the dose level; the quantity of interest—acid chloride delivered to the tumor—is governed by the same resistance fraction ϕ .

7 Remarks and scope

Remark 3 (Nonlinear embolization). *Proposition 1* requires a linear two-port bridge. Once embolization-driven viscosity feedback engages (as in *Onyx/Lipiodol* models), k_{br} becomes state-dependent and (11) holds only in the linearized/local sense. For the pre-embolization flow-field question the two-port reduction and the equivalence are exact; for the delivered-dose-with-embolization question they furnish the leading-order term plus a state-dependent correction to $k_{\text{br}}(t)$.

Remark 4 (Few generations, exhaustive enumeration). *In thermoembolization the injection sits within one to three branch generations of the tumor along a near-straight path, so the set of physically relevant break configurations—location, width, length across at most three generations—is small enough to enumerate exhaustively rather than sample, and each is scored by its ϕ from Corollary 1.*

Remark 5 (Reconstruction operator caveat). *Whether a forward imaging operator that generates realistic breaks is a slab mean or a maximum-intensity projection determines which breaks appear: a mean pulls thin vessels below threshold (partial-volume breaks), whereas a MIP retains the brightest voxel and suppresses them, leaving only genuinely intensity-dead gaps. The dominant break mechanism—and thus the operator used to synthesize ground-truth breaks—should be fixed by comparing, on a large proximal artery, whether the thick-slab HU matches the mean or the maximum of the thin-slice HU.*